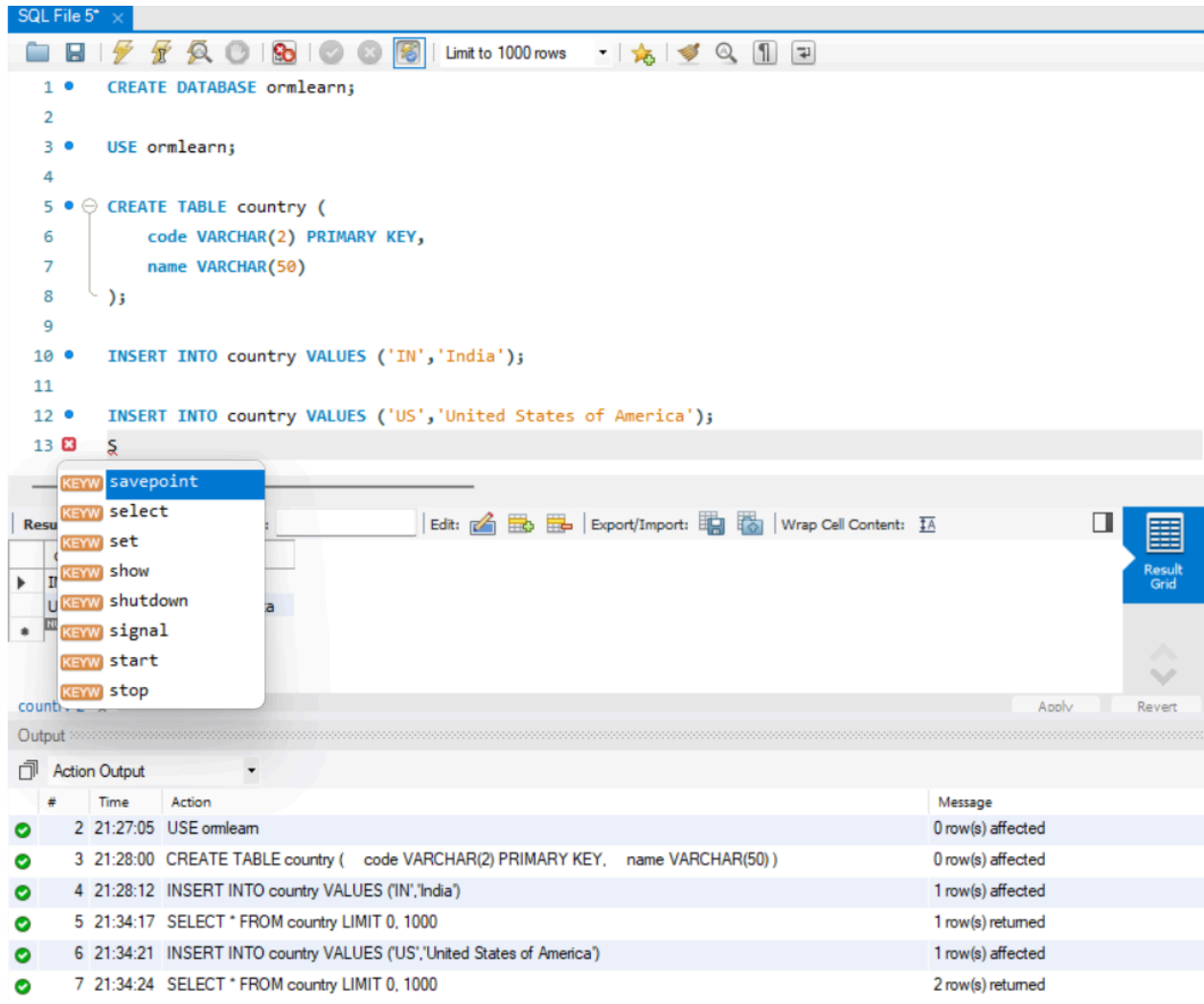


Hands on 1

Spring Data JPA - Quick Example

MySQL WorkBench:



The screenshot shows the MySQL Workbench interface with a SQL script being executed. The script contains the following statements:

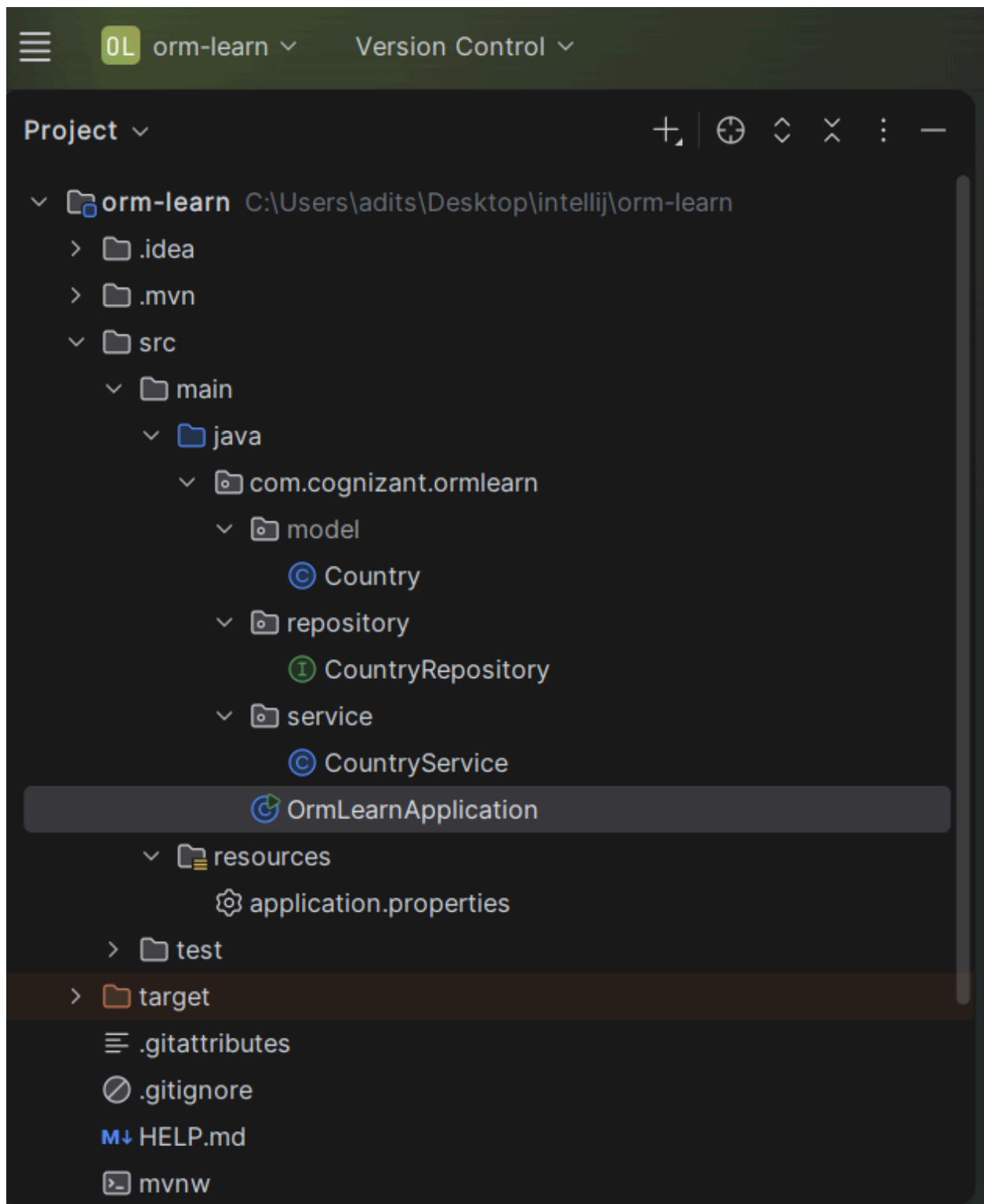
```
1 • CREATE DATABASE ormllearn;
2
3 • USE ormllearn;
4
5 • CREATE TABLE country (
6     code VARCHAR(2) PRIMARY KEY,
7     name VARCHAR(50)
8 );
9
10 • INSERT INTO country VALUES ('IN','India');
11
12 • INSERT INTO country VALUES ('US','United States of America');
13
```

A context menu is open over the script, showing options: savepoint, select, set, show, shutdown, signal, start, and stop.

The Output pane at the bottom shows the execution results:

#	Time	Action	Message
✓ 2	21:27:05	USE ormllearn	0 row(s) affected
✓ 3	21:28:00	CREATE TABLE country (code VARCHAR(2) PRIMARY KEY, name VARCHAR(50))	0 row(s) affected
✓ 4	21:28:12	INSERT INTO country VALUES ('IN','India')	1 row(s) affected
✓ 5	21:34:17	SELECT * FROM country LIMIT 0, 1000	1 row(s) returned
✓ 6	21:34:21	INSERT INTO country VALUES ('US','United States of America')	1 row(s) affected
✓ 7	21:34:24	SELECT * FROM country LIMIT 0, 1000	2 row(s) returned

File Structure:



Country.java

```

package com.cognizant.ormlearn;

import java.util.List;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.ApplicationContext;

import com.cognizant.ormlearn.model.Country;
import com.cognizant.ormlearn.service.CountryService;

@SpringBootApplication
public class OrmLearnApplication {

    private static final Logger LOGGER =
        LoggerFactory.getLogger(OrmLearnApplication.class);

    private static CountryService countryService;

    public static void main(String[] args) {

        ApplicationContext context =
            SpringApplication.run(OrmLearnApplication.class, args);

        LOGGER.info("Inside main");

        countryService = context.getBean(CountryService.class);

        testGetAllCountries();
    }

    private static void testGetAllCountries() {

        LOGGER.info("Start");

        List<Country> countries = countryService.getAllCountries();

        LOGGER.debug("countries={}", countries);

        LOGGER.info("End");
    }
}

```

CountryRepository.java

```

package com.cognizant.ormlearn.repository;

import org.springframework.data.jpa.repository.JpaRepository;
import org.springframework.stereotype.Repository;

import com.cognizant.ormlearn.model.Country;

@Repository
public interface CountryRepository extends JpaRepository<Country, String> {

}

```

CountryService.java

```

package com.cognizant.ormlearn.service;

import java.util.List;

import org.springframework.beans.factory.annotation.Autowired;
import org.springframework.stereotype.Service;

import com.cognizant.ormlearn.model.Country;
import com.cognizant.ormlearn.repository.CountryRepository;

import jakarta.transaction.Transactional;

@Service
public class CountryService {

    @Autowired
    private CountryRepository countryRepository;

    @Transactional
    public List<Country> getAllCountries() {
        return countryRepository.findAll();
    }

}

```

OrmLearnApplication.java

```

package com.cognizant.ormlearn;

import java.util.List;

import org.slf4j.Logger;
import org.slf4j.LoggerFactory;

import org.springframework.boot.SpringApplication;
import org.springframework.boot.autoconfigure.SpringBootApplication;
import org.springframework.context.ApplicationContext;

import com.cognizant.ormlearn.model.Country;
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@SpringBootApplication
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        LoggerFactory.getLogger(OrmLearnApplication.class);

    private static CountryService countryService;

    public static void main(String[] args) {

        ApplicationContext context =
            SpringApplication.run(OrmLearnApplication.class, args);

        LOGGER.info("Inside main");

        countryService = context.getBean(CountryService.class);

        testGetAllCountries();
    }

    private static void testGetAllCountries() {

        LOGGER.info("Start");

        List<Country> countries = countryService.getAllCountries();

        LOGGER.debug("countries={}", countries);

        LOGGER.info("End");
    }
}

```

Output:

The image shows a screenshot of an IDE with the 'Run' tab selected. The top toolbar includes icons for 'orm-learn', 'Version Control', 'Current File', and 'Unlock Ultimate'. The 'Run' tab shows the 'OrmLearnApplication' class. The console output displays various log messages, including warnings about deprecation and connection info, and status messages for the application's lifecycle. The bottom status bar shows the file path 'orm-learn > src > main > java > com > cognizant > orm-learn > OrmLearnApplication' and the file size '17:1 (1137 chars, 44 line breaks)'.

```
WARN o.h.orm.deprecation constructDialect 152 HHH90000025: MySQLDialect does not need to be specified explicitly using 'hibernate.dialect' (remove the prope
INFO o.h.o.connections.pooling logConnectionInfo 179 HHH10001005: Database info:
[host:3306/ormLearn]

afined

fault REPEATABLE_READ]

INFO org.hibernate.orm.core initiateService 53 HHH000489: No JTA platform available (set 'hibernate.transaction.jta.platform' to enable JTA platform integrat
INFO rEntityManagerFactoryBean buildNativeEntityManagerFactory 464 Initialized JPA EntityManagerFactory for persistence unit 'default'
INFO .j.QueryEnhancerFactories <cInit> 48 Hibernate is in classpath; If applicable, HQL parser will be used.
INFO c.c.o.OrmLearnApplication logStarted 60 Started OrmLearnApplication in 3.467 seconds (process running for 4.005)
INFO c.c.o.OrmLearnApplication main 28 Inside main
INFO c.c.o.OrmLearnApplication testGetAllCountries 37 Start
DEBUG org.hibernate.SQL logStatement 126 select c1_0.code,c1_0.name from country c1_0
DEBUG c.c.o.OrmLearnApplication testGetAllCountries 41 countries=[Country{code='IN', name='India'}, Country{code='US', name='United States of America'}]
INFO c.c.o.OrmLearnApplication testGetAllCountries 43 End
ok INFO rEntityManagerFactoryBean destroy 693 Closing JPA EntityManagerFactory for persistence unit 'default'
ok INFO c.z.h.HikariDataSource close 349 HikariPool-1 - Shutdown initiated...
ok INFO c.z.h.HikariDataSource close 351 HikariPool-1 - Shutdown completed.
```